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Portfolio Regulation of Financial Institutions with Market Power

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1 Big picture

- **Question.** When financial institutions have market power and trade strategically with price impact, does giving them access to a richer set of securities necessarily improve risk sharing, or can portfolio restrictions actually raise welfare?
- **Setting.** The paper studies a two-date endowment economy with a competitive fringe and a finite number of large strategic investors. The regulator can restrict the *asset span*, that is, the set of risks investors are allowed to trade.
- **Core friction.** Strategic investors distort trades to move prices in their favor. Buyers ration demand to push prices down, sellers ration supply to push prices up. This generates unrealized gains from trade even though the underlying risks are technologically tradeable.
- **Main theoretical insight.** Complete markets are not generally efficient once investors have market power. In fact, when the number of strategic investor types is smaller than the number of states, unrestricted access to the full Arrow-security menu is *constrained inefficient*. Judicious forms of market incompleteness can improve welfare.
- **Why incompleteness can help.** Complete markets let investors exploit price impact in a fully state-contingent way. Bundling securities across states forces distortions on one margin to spill over to others, which can mute rent extraction and improve overall risk sharing.
- **Application.** The paper applies the theory to sovereign credit default swaps (CDS). A ban on “naked” CDS positions — forcing investors to hold CDS only together with the underlying bond — can improve overall risk sharing even if some narrow liquidity measures, such as the bond bid-ask spread, deteriorate.
- **Why this paper matters.** It changes the usual benchmark for financial regulation. The standard view is that more complete markets are better. This paper shows that with strategic traders, the relevant trade-off is between *gains from trade* and *rent-seeking through price impact*. Cross-asset spillovers are therefore central for policy evaluation.

2 Data and measurement (key objects)

2.1 Agents, endowments, and state space

- **Dates and states.** There are two dates, $t \in \{1, 2\}$, and a finite set of states $z \in Z$ realized at date 2 with probabilities $\pi(z)$.
- **Strategic investors.** There are N types of strategic agents, indexed by $i \in \{1, \dots, N\}$. An agent of type i receives date-1 wealth $\mu_i w_i$ and state-contingent date-2 endowment $\mu_i y_i(z)$.
- **Competitive fringe.** A competitive fringe of mass m_f takes prices as given and prices all securities through its marginal utility. It is analytically convenient because the pricing functional can be written in closed form.
- **Market power parameter.** The size parameter μ_i measures how much wealth and income each individual strategic agent controls, and therefore how much price impact it has. The competitive benchmark is the limiting case $\mu_i = 0$ for all i .

- **Preferences.** Strategic agents have homothetic and concave utility over consumption at both dates. The fringe has quasi-linear preferences in date-1 consumption and concave utility over date-2 consumption.

2.2 Financial markets and the asset span

- **Securities.** There are $J \leq Z$ securities with payoff vectors $\{x_j(z)\}_{j=1}^J$.
- **Asset span.** A market structure is summarized by the payoff matrix

$$X \in \mathbb{R}^{J \times Z},$$

where the j th row gives the state-contingent payoff of security j .

- **Complete versus restricted markets.** The market structure is complete if

$$\text{rank}(X) = Z,$$

and restricted if

$$\text{rank}(X) < Z.$$

- **Portfolio positions.** Type i holds $a_{i,j}$ units of asset j , with $a_{i,j} < 0$ denoting a short position. The fringe holds $a_{f,j}$. Market clearing is

$$\sum_{i=1}^N a_{i,j} + m_f a_{f,j} = 0 \quad \text{for all } j.$$

2.3 Unobservables, pricing, and timing

- **Unobserved pricing object.** The key endogenous object is each investor's state-price valuation, or stochastic discount factor (SDF),

$$m_i(z) \equiv \frac{u'_{i,2}(c_{2,i}(z))}{u'_{i,1}(c_{1,i})}.$$

- **Timing.** At date 1, strategic agents choose consumption and portfolios taking as given the other strategic investors' demand schedules, but internalizing how their own trade moves equilibrium prices through the fringe. At date 2, state z realizes and assets pay off.
- **Main friction.** Because prices depend on the entire vector of strategic positions, each strategic investor internalizes price impact. This is what creates wedges between private marginal valuations and market prices.

2.4 Empirical application and calibration targets

- **Application.** The empirical application studies sovereign bond and sovereign CDS markets.
- **Two strategic types.** In the calibration, the two strategic types are *clients* and *dealers*. Dealers end up selling bonds and CDS to clients.

- **Data moments.** The model is matched to four moments: the mean sovereign bond yield, the mean sovereign bond bid-ask spread, the mean sovereign CDS bid-ask spread, and the CDS-bond basis.
- **Sample behind the moments.** The moments are drawn from global sovereign bond and CDS markets and are intended to capture broad patterns rather than a single country episode.

3 Models and empirical specifications (all equations + interpretation)

3.1 Strategic portfolio choice and Cournot-Walras equilibrium

A representative strategic investor of type i solves

$$U_i = \max_{c_{1,i}, \{a_{i,j}\}_{j=1}^J} \left\{ u_{i,1}(c_{1,i}) + \sum_{z \in Z} \pi(z) u_{i,2}(c_{2,i}(z)) \right\}$$

subject to

$$\begin{aligned} \mu_i c_{1,i} &= \mu_i w_i - \mu_i \sum_j P_j(A) a_{i,j}, \\ \mu_i c_{2,i}(z) &= \mu_i y_i(z) + \mu_i \sum_j x_j(z) a_{i,j}. \end{aligned}$$

The competitive fringe solves the same portfolio problem but takes prices as given.

- **Interpretation.** This is a Cournot-Walras equilibrium: strategic investors choose quantities knowing that quantities move prices, while the fringe is competitive and enforces no arbitrage.
- **Why this matters.** The model preserves general-equilibrium pricing discipline but allows large institutions to behave strategically.

The paper's first major result is a characterization of equilibrium prices and price impact. Writing the diagonal matrix of objective probabilities as

$$\Pi \equiv \text{diag}(\pi(z)),$$

Proposition 1 implies:

$$P_j(A) = p_j(A) = \sum_{z \in Z} \pi(z) x_j(z) m_f(z),$$

where the fringe SDF is

$$m_f(z) = u'_f(c_{2,f}(z)).$$

The representative type- i price-impact matrix is

$$\mathbf{3}(A) = \frac{\mu}{m_f} X \Theta X', \quad \Theta_{zz} = -m'_f(z) \geq 0,$$

and the strategic investor's first-order condition is

$$X \Pi M_i = p(A) + \frac{\mu}{m_f} X \Theta X' a_i.$$

- **Interpretation.** Asset prices are pinned down by the fringe’s marginal utility, and strategic distortions enter through the extra price-impact term on the right-hand side.
- **Equivalent asset spans.** Consumption allocations are invariant to trading alternative sets of securities with the same span. This turns out to be crucial for the paper’s “missing asset” logic later on.

3.2 Complete markets, Arrow securities, and privately optimal distortions

To isolate the nature of distortions, the paper studies the complete-markets benchmark in which the asset menu is the full set of Arrow securities. Let $p(z)$ be the price of the Arrow claim paying in state z . Proposition 3 shows that

$$p(z) = \pi(z)m_f(z),$$

and the type- i optimal holding of Arrow security z satisfies

$$\pi(z)m_i(z) = p(z) + \frac{\mu}{m_f}p'(z)a_i(z).$$

- **Interpretation.** Under perfect competition, private state prices $\pi(z)m_i(z)$ would equal market prices. With price impact, they differ by an endogenous wedge proportional to position size.
- **Economic meaning.** Buyers choose to buy too little and sellers choose to sell too little relative to the first best because both want price concessions.
- **Lost gains from trade.** Risk sharing is inefficient not because some risks are technologically unavailable, but because strategic traders ration trade.

The paper’s first canonical example considers *pure diversifiable risk*: two ex ante symmetric strategic types, one rich in state 1 and poor in state 2, the other the reverse. Let perfect risk sharing require selling one claim in the high-income state and buying one claim in the low-income state. Under price impact, the equilibrium positions become

$$a^S = -1 + \delta^S, \quad a^B = 1 - \delta^B,$$

with

$$\delta^S > \delta^B > 0.$$

- **Interpretation.** Sales are distorted more than purchases. The reason is that agents sell when they are rich, so marginal utility is flatter and the private cost of distorting trade is lower.
- **Illiquidity externality.** Because the pricing functional is nonlinear, one investor’s rationing changes price impact for other investors. Strategic distortions therefore create an illiquidity externality in addition to direct rent extraction.

3.3 Counterfactual incomplete markets and the paper’s “missing asset” idea

The paper’s second big step is to reinterpret the complete-markets equilibrium with strategic trade as if it had been generated by *competitive trade in an incomplete market*. The idea is to hold fixed the equilibrium consumption allocation and ask: what is the largest restricted asset span such that, if all

agents traded competitively under that span, no one would want to trade away from the distorted allocation?

Formally, Proposition 4 constructs a return matrix $\tilde{X}$ of maximal rank $K < Z$ whose columns satisfy

$$M\Pi\tilde{x}_k = \iota_N \quad \text{for all } k,$$

where M stacks the strategic agents' SDF vectors row by row and ι_N is a vector of ones.

- **Interpretation.** The implied incomplete market is the richest asset span that all strategic agents would price competitively in a way consistent with the distorted complete-markets allocation.
- **Key implication.** The risks that are *outside* this span are exactly the risks that strategic price impact leaves poorly shared. Those are the paper's "missing assets."
- **Why this is useful.** Instead of describing distortions security by security, the paper describes them directly in terms of the *risk exposures* that remain rationed in equilibrium.

The two main examples are especially intuitive.

Example 2: pure diversifiable risk. In the first canonical setting, the implied counterfactual asset span is simply a risk-free bond. The missing asset is the idiosyncratic swap with payoffs

$$[-1, 1].$$

- **Interpretation.** The economy behaves as if the only tradeable asset were riskless, so the diversifiable risk swap is the asset that strategic traders are effectively withholding.

Example 3: aggregate risk with heterogeneous preferences. In the second canonical setting, one strategic type is risk-neutral and the other risk-averse, and all aggregate risk is common. The implied restricted asset span is a *levered market index*. The missing asset is an aggregate risk swap with payoffs proportional to

$$[m_{rn} - m_h, m_{rn} - m_l],$$

which pays in the high state and loses in the bad state.

- **Interpretation.** Even though aggregate risk is tradeable, the specific insurance trade that would best transfer bad-state resources is rationed by strategic behavior.

3.4 Constrained efficiency and why complete markets can be too much of a good thing

The regulator chooses the market structure X and can also use budget-balanced date-1 transfers. The paper defines an asset span as *constrained efficient* if the planner cannot generate a Pareto improvement by changing the span and introducing such transfers.

The key necessary condition for a constrained efficient asset span is Proposition 5:

$$\sum_{i=1}^N \Pi(M_i - M_f) \otimes a_i + \frac{\mu}{m_f} \nabla_{\tilde{X}_J} [a'_i X \Theta X' a_i] = \mathbf{0}_{ZJ}.$$

The paper interprets this as a decomposition into two channels:

$$\begin{aligned} & \text{Change in risk-sharing efficiency} \\ & + \text{Change in price-impact rents} = 0. \end{aligned}$$

Moreover, if

$$N < Z,$$

complete markets are constrained inefficient.

- **Interpretation of the first term.** Changing the asset span changes which consumption profiles can be traded, and therefore which gains from trade are realizable.
- **Interpretation of the second term.** Changing the asset span also changes how much state-contingent market power investors can exploit.
- **Main message.** The planner trades off more risk-sharing opportunities against more room for strategic rent extraction. Once strategic distortions are present, completing markets is not automatically desirable.

Proposition 6 then delivers explicit Pareto-improving restrictions in the canonical examples.

- **Setting 1.** If $u'(\bar{y}) \geq 1$, restricting trade to the swap with payoffs

$$[1, -1]$$

generates a Pareto improvement over unrestricted complete markets.

- **Setting 2.** In the heterogeneous-preferences case, there exists a swap with payoffs

$$[-1, x_l], \quad x_l > 0,$$

such that restricting agents to trade only this swap achieves a Pareto improvement over unrestricted complete markets.

- **Economic intuition.** The optimal bundle forces the investor to distort multiple state contingencies at once. That bundled distortion is less attractive than distorting each Arrow security separately.

3.5 Application to sovereign bond and CDS regulation

The application specializes the model to two states, two strategic types (clients and dealers), and a competitive fringe, all with log utility.

Complete markets. There are two securities:

- a risky sovereign bond with face value B that pays B in state 1 and 0 in state 2,
- a CDS contract that pays 0 in state 1 and B in state 2.

Together these span the two-state economy.

Regulated market. Under a ban on naked CDS positions, agents can trade only a *covered bond* that pays

$$[B, \rho B].$$

This is equivalent to forcing bond and CDS positions to be held in a fixed proportion.

- **Interpretation.** The regulation is an asset-bundling restriction. It removes one degree of freedom in risk trading and therefore directly maps into the paper’s theory of constrained-efficient asset spans.

The calibration uses four target moments under the complete-markets benchmark:

- bond yield: 5.08%,
- bond bid-ask spread: 1.00% of par,
- CDS bid-ask spread: 13.47% of midpoint,
- CDS-bond basis: -1.00% .

The associated calibrated parameters are

$$Y = 2, \quad B = 0.05, \quad \pi = 0.048, \quad \frac{\mu}{m_f} = 0.133, \quad \delta = 0.994, \quad \rho B = 0.6917.$$

- **Identification logic inside the calibration.** Default probability mostly disciplines the bond yield; relative strategic size disciplines price impact through bid-ask spreads; the aggregate shock δ helps pin down the CDS spread; and the CDS-bond basis pins down endowment dispersion.

3.6 Partial regulation and broader applications

The paper then studies what happens if regulation is not market-wide.

Example 4: imperfect coverage in the pure diversifiable-risk case. If only a fraction of investors are forced to trade the swap, price impact still falls because constrained investors distort less and their offsetting positions improve fringe consumption smoothness.

Example 5: asymmetric regulation with aggregate risk. If only one investor type is constrained, regulation can redistribute market power toward unconstrained institutions. Liquidity improves in some markets and worsens in others, and surplus shifts across investor types.

- **Interpretation.** Market-wide regulation is cleaner. Partial coverage can simply move rent extraction from regulated institutions to unregulated ones.
- **Other applications.** The paper argues the same logic may apply to capital guarantee products and variable annuities, where structured products bundle market exposure with downside insurance and are often supplied in concentrated, illiquid markets.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard intuition

- In standard complete-markets theory with competitive agents, more securities are always weakly better because they expand the feasible set of allocations.
- Here that logic fails because securities do not just allocate risk — they also determine *how finely investors can target their market-power distortions*.
- Therefore, security design and portfolio regulation are part of the strategic environment, not a neutral background assumption.

4.2 What identifies the constrained-efficient asset span

- The paper identifies the relevant missing risks from the cross-section of strategic agents' state prices.
- Proposition 4 shows how to back out a competitive incomplete-market span that rationalizes the same distorted allocation as the complete-markets strategic equilibrium.
- This is the paper's bridge from *observed strategic distortions* to *regulatory design*: the missing asset in the counterfactual economy is the asset whose bundling can restore risk sharing.
- The sufficient condition

$$N < Z$$

is crucial. With fewer strategic types than states, complete markets create too many state-contingent degrees of freedom for strategic rent extraction.

4.3 What pins down the CDS calibration

- The application is not identified by reduced-form causal variation; it is identified by matching equilibrium moments to a fully specified model.
- The discipline comes from using both asset prices and liquidity measures. That is important because the same policy can lower one spread and raise another.
- The calibration also makes a strong modeling assumption: the regulated covered bond is presumed to be reasonably aligned with the true gains from trade under perfect competition.
- Appendix Figure C1 shows why this assumption matters. If the bundling parameter ρ is too far from the value consistent with underlying trading needs, the regulation can become welfare-reducing.

4.4 Relation to the literature

- **Relative to second-best incomplete markets.** The classic literature shows that complete markets are first-best under competitive trade. This paper shows that with strategic trade and price impact, incomplete markets can dominate complete markets.

- **Relative to Cournot-Walras strategic trade.** The paper extends that tradition to a setting where the set of tradeable portfolios is itself a policy object.
- **Relative to market design and financial innovation.** The paper is closely related to work on decentralized exchange, security design, and endogenous incompleteness, but emphasizes bundling as a way to attenuate market power in risk sharing.
- **Relative to recent intermediary asset-pricing work.** The policy application connects to modern debates about concentrated dealer markets, CDS-bond interactions, and the supply of structured insurance products.

5 Tables and figures

5.1 Table 1: Calibration targets and associated parameters

Read:

- This table explains how the CDS application is disciplined.
- The model is pinned down by four empirical moments from bond and CDS markets: yield, bond bid-ask spread, CDS bid-ask spread, and the CDS-bond basis.
- The calibrated relative size of strategic agents, $\mu/m_f = 0.133$, is the key object controlling the strength of price impact.
- The bundling parameter $\rho B = 0.6917$ is not just a technical detail; it governs how the covered-position regulation maps risk across states.

Table 1
Calibration targets and associated parameters. Y and B fix the scale of the economy and thus have no direct counterparts in the data. All market participants have log utility

Data moment	Value	Parameter	Calibrated value
Aggregate steady state endowment	—	Y	2
Face value of bond	—	B	0.05
Bond yield (%)	5.08	Default prob. π	0.048
Bid-ask spread for bonds (% of par)	1.00	Relative size $\frac{\mu}{m_f}$	0.133
Bid-ask spread for CDS (% of mid-price)	13.47	Agg. shock δ	0.994
CDS-bond basis (%)	−1.00	Dispersion ρB	0.6917

Table 2
Outcomes for complete markets benchmark and regulated markets with covered bond only. The equivalent variation is computed as a wealth change in the complete markets economy, holding prices and price impacts fixed. The wealth transfer required for a Pareto improvement is a date-1 transfer from clients to dealers, and it is measured by recomputing all equilibrium outcomes given the transfer. It is reported relative to initial client wealth

Data moment	Complete markets	Covered bond only
Bond yield (%)	5.08	5.05
Bid-ask spread for bonds (% of par)	1.00	1.32
Bid-ask spread for CDS (% of mid-price)	13.47	8.29
CDS-bond basis (%)	−1.00	−0.56
Client equivalent variation relative to initial wealth (bps)	—	0.47
Dealer equivalent variation relative to initial wealth (bps)	—	−0.33
Wealth transfer for Pareto improvement (bps)	—	0.3–0.46

5.2 Table 2: Complete markets benchmark versus covered-bond-only regulation

Read:

- The regulation lowers sovereign borrowing costs and sharply improves CDS-market liquidity.
- The table reports bond yield changing from 5.08% to 5.05%, bond bid-ask spread from 1.00% to 1.32%, CDS bid-ask spread from 13.47% to 8.29%, and the CDS-bond basis from −1.00% to −0.56%.
- The policy is *distributional*: clients gain the equivalent of 0.47 basis points of initial wealth, dealers lose 0.33 basis points.
- A transfer of roughly 0.3–0.46 basis points of initial client wealth to dealers would make the policy a Pareto improvement.

- The key lesson is not that every asset-specific liquidity statistic improves. It is that general-equilibrium welfare can improve even when the bond bid-ask spread worsens, because cross-asset spillovers matter.

5.3 Appendix Figure C1: what happens when regulation is mismatched to gains from trade

Read:

- The figure varies the covered-bond payoff parameter ρB while holding the rest of the calibration fixed.
- Bond yields and bond bid-ask spreads rise with ρB , while the CDS bid-ask spread falls.
- Client welfare is hump-shaped, and utilitarian welfare can become negative if ρB is chosen too high or too low.
- This figure is important because it shows that the paper is *not* saying “any bundling helps.” The bundling must be aligned with the actual gains from trade.

6 Short summary

- **Conclusion.** With market power in financial markets, unrestricted complete markets generally permit too much state-contingent rent extraction. Portfolio restrictions can therefore improve risk sharing by bundling risks in ways that reduce strategic manipulation.
- **Key mechanism.** The regulator changes the asset span, which changes both the set of feasible consumption transfers and the strategic scope for exploiting price impact.
- **Application result.** In the CDS application, banning naked sovereign CDS positions can improve overall risk sharing even if it raises the bond bid-ask spread, because it lowers the CDS spread and reduces the CDS-bond basis.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that complete markets are not the correct efficiency benchmark once financial institutions have market power.
- It provides a general equilibrium framework in which the regulator can evaluate portfolio restrictions as changes in the asset span.
- The paper shows that, under a simple and economically intuitive condition, unrestricted complete markets are constrained inefficient, and certain forms of incompleteness are constrained efficient.

7.2 Economic intuition

- A strategic trader values state-contingent flexibility because it can choose exactly where to ration supply or demand to obtain price concessions.
- If the regulator bundles states together inside a composite asset, the trader loses that flexibility. Distorting one state now forces distortion of another state too.
- If the bundle is designed to line up with the economy's true gains from trade, the reduction in rent-seeking dominates the lost flexibility from incompleteness.

7.3 Takeaway

This paper is best understood as a theory of *security design under market power*. Its central message is that the welfare effect of portfolio regulation cannot be read off from a single market's price or bid-ask spread. The regulator must account for how portfolio constraints reshape trading incentives across the entire asset span. In concentrated markets, the efficient policy problem is therefore not “how do we complete markets?” but rather “how do we bundle risks so that investors can still trade what matters while having less room to exploit price impact?”